

Explainable Spectro-Temporal Modeling for Indian Raga Classification Using Hybrid Neural Network

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Abstract

Indian classical music, particularly the Hindustani tradition, is renowned for its rich melodic structures, emotional expressiveness, and deep cultural roots. Within this musical framework, ragas serve as foundational melodic templates that guide improvisation and composition. However, the intricate tonal architectures, flexible temporal progressions, and subtle ornamentations (gamakas, meends) that define ragas pose significant challenges for computational classification and automated analysis. These challenges are compounded by the inherent subjectivity and interpretive nature of Indian classical performances, where the same raga can be rendered differently by different artists.

To address this complexity, we propose a robust deep learning-based approach that simultaneously leverages both spectral and temporal features for accurate raga identification. In this study, we introduce a hybrid architecture that combines Convolutional Neural Networks (CNNs) for spatial pattern recognition and Long Short-Term Memory (LSTM) networks for capturing long-range temporal dependencies. The input features are based on Mel-Frequency Cepstral Coefficients (MFCCs), which effectively model the perceptual aspects of audio signals and serve as a compact, information-rich representation of the time-frequency domain.

To improve transparency and trust in the model’s predictions, we incorporate SHapley Additive exPlanations (SHAP), a widely adopted explainable AI (XAI) technique. SHAP enables both global and local interpretability by attributing prediction outcomes to specific MFCC features, revealing how different spectral characteristics influence the classification decision. This is particularly important in domains like music education, archival studies, and musicological analysis, where understanding why a model made a decision can be as critical as the decision itself.

Our model was trained on a curated dataset comprising labeled audio samples from ten well-known Hindustani ragas. The CNN-LSTM network achieved an impressive overall accuracy of 95.56% on the test set, with macro and weighted F1-scores exceeding 0.95. SHAP-based visualizations revealed that specific MFCC coefficients (e.g., MFCC-39, MFCC-32) consistently influenced the classification across multiple raga categories. These insights validate the effectiveness of both the feature extraction and model architecture.

In conclusion, this research presents a high-performing, interpretable deep learning pipeline for Indian classical raga classification. By integrating the strengths of CNNs, LSTMs, and SHAP-based explainability, our approach offers a transparent, scalable, and culturally sensitive solution with meaningful applications in digital heritage preservation, computational ethnomusicology, and intelligent music education systems.

Introduction

Indian classical music is regarded as one of the most profound and spiritually rich musical traditions in the world. Rooted in centuries of cultural and philosophical heritage, it is known for its intricate melodic structures, rhythmic precision, and deep emotional expressiveness. At the heart of this tradition lies the concept of the raga—a sophisticated melodic framework that serves as the foundation for both improvisation and composition. A raga is not merely a scale but a complex set of rules and characteristics that govern the use of notes (swaras), ornamentations (gamakas), tonal relationships (vadi-samvadi), and contextual elements such as time-of-day or seasonal associations.

Each raga possesses a unique identity defined by its aroha (ascending scale), avaroha (descending scale), pakad (signature phrases), and emotive quality (rasa). While the theoretical aspects of ragas are codified, the performance practice is inherently fluid and expressive, with significant room for improvisation. This blend of strict structure and interpretive freedom introduces substantial variability, making the task of raga identification particularly challenging—especially in live or semi-improvised recordings.

Manual recognition of ragas has historically depended on expert human listeners with years of dedicated training. Even seasoned musicians and scholars may face difficulty distinguishing between ragas that share similar note structures or stylistic features. For example, ragas like Bageshree and Rageshree, or Darbari and Adana, often feature overlapping melodic contours, making precise classification a non-trivial task.

The digital age has witnessed an explosion in the availability of Indian classical music through online platforms, streaming services, and archival initiatives. However, the tools to analyze and index this content have not kept pace with its volume. As a result, vast repositories of raga-based music remain underutilized due to the absence of automated systems capable of identifying and tagging ragas at scale.

Despite advancements in music information retrieval (MIR) and artificial intelligence (AI), most existing computational models are optimized for Western music structures and fall short when applied to the nuanced temporal and tonal dynamics of Indian classical traditions. Techniques such as pitch tracking, onset detection, or frequency-based segmentation often break down in the face of microtonal transitions, continuous pitch glides (meend), and improvisational deviations from canonical forms.

This scenario highlights the urgent need for intelligent, data-driven systems capable of understanding not only the pitch content but also the temporal evolution and expressive nuances of raga performance. A system that can effectively model these subtleties would offer transformative applications across music education, digital archiving, and interactive musicology. Such a system must balance performance accuracy with interpretability, allowing musicians and researchers to trust, analyze, and verify the results.

In this context, deep learning architectures—especially those that incorporate both spatial and temporal modeling—hold significant promise. When paired with explainability techniques such as SHAP, these models can not only predict the raga with high confidence but also provide insights into why a particular prediction was made. This intersection of performance and interpretability is key to bringing computational music analysis in Indian classical music into the next era of technological and cultural relevance.

Literature Survey

Recent advancements in audio signal processing and deep learning techniques have significantly accelerated research in the computational classification of Indian classical and related musical genres. These technological breakthroughs have made it increasingly possible to explore the rich and complex musical traditions of India through automated systems. Among the early contributions, Jothilakshmi and Kathiresan [1] employed Mel-Frequency Cepstral Coefficients (MFCCs) in conjunction with Gaussian Mixture Models (GMMs) to develop automatic systems for genre classification. While their work was foundational and offered meaningful insights into feature-based music analysis, it focused primarily on broader musical genres and did not attempt the more complex task of identifying individual ragas, which are central to Indian classical music. In a related study, Ujlambkar and Attar [2] utilized Support Vector Machines (SVMs) to classify Indian popular music based on mood. Their approach was centered on emotional tone recognition, which, although relevant, did not incorporate the structured melodic and temporal intricacies that define classical ragas. The study therefore lacked applicability to raga-level classification tasks where the structural identity of a raga—including its note sequences, microtones, and ornamentations—is crucial. Building on the idea of temporal modeling, Roy et al. [3] introduced a raga-based recommendation system that employed Recurrent Neural Networks (RNNs) to capture time-dependent characteristics of musical performance. While the use of RNNs allowed for modeling sequential data—a key requirement for understanding raga dynamics—their system was primarily built for recommendation rather than classification, and therefore did not achieve the granularity required for accurate raga identification. Waghmare and Sonkamble [4] explored classical machine learning approaches such as K-Nearest Neighbors (KNN) and Naive Bayes classifiers for raga classification. However, these models relied heavily on handcrafted features, which are often less adaptable and do not generalize well across diverse raga structures or performance styles. This limitation restricts the flexibility and scalability of such systems when applied to varied or unseen datasets. A significant advancement came from the work of Bisht et al. [5], who combined MFCC feature extraction with a Recurrent Neural Network-Long Short-Term Memory (RNN-LSTM) architecture. Their model was effective in recognizing temporal dependencies and showed strong results in Hindustani music genre classification. However, it still lacked specificity at the raga level and did not incorporate interpretability techniques to explain the model’s predictions. Singha et al. [6] leveraged Convolutional Neural Networks (CNNs) to classify Indian classical ragas, demonstrating that deep learning architectures can effectively learn melodic structures. Nonetheless, their approach was largely a black-box system, providing little transparency or explanation of how decisions were made by the model. Overall, while these studies have advanced the state of computational music analysis, they either concentrated on broader categories like genre and mood, or lacked integration with explainable AI (XAI) methodologies. This underscores the pressing need for more interpretable and high-performing models that can provide insight into how predictions are made—especially for the nuanced task of Indian classical raga classification.

Research Gap

Despite the growing interest in computational musicology and the application of machine learning to Indian classical music, several critical research gaps persist—particularly in the context of raga identification, which is more complex and nuanced than general genre or mood classification.

1. **Limited Focus on Raga-Level Classification:** Much of the existing literature, including works by Jothilakshmi and Kathiresan [1], and Ujlambkar and Attar [2], has predominantly focused on genre- or mood-level classification rather than the identification of individual ragas. These broader categorizations ignore the intricate structural and melodic rules (such as Arohana, Avarohana, Pakad, Vadi-Samvadi) that define ragas, leading to models that lack the granularity required for fine-level raga recognition.
2. **Reliance on Traditional Machine Learning Methods:** Earlier models such as those using Gaussian Mixture Models (GMMs), Support Vector Machines (SVMs), K-Nearest Neighbors (KNN) and Naive Bayes have heavily relied on handcrafted features like MFCCs. While useful, these features may not capture the contextual, temporal, and expressive richness inherent in raga performances. Moreover, handcrafted features are prone to dataset-specific bias and lack the capacity to generalize across diverse raga forms or performance styles.
3. **Black-Box Deep Learning Approaches with No Interpretability:** While recent works like those of Roy et al. [3], Bisht et al. [5], and Singha et al. [6] have introduced deep learning architectures (RNNs, LSTMs, CNNs), their models are largely treated as black boxes. These approaches fail to provide any explanation or rationale behind the classification decisions, which poses a significant barrier in domains like classical music, where cultural and subjective interpretations are integral. The lack of explainable AI (XAI) techniques in these models restricts their usability in educational and archival settings where transparency and interpretability are essential.
4. **Insufficient Temporal and Melodic Modeling:** Although some models attempt to capture temporal patterns (e.g., via RNNs or LSTMs), they fall short of learning long-term melodic dependencies or structural hierarchies that define a raga. Additionally, CNN-based models such as those used by Singha et al. have been limited in scope, as they mostly capture local spectral patterns, not the sequential melodic flow necessary for distinguishing closely related ragas.
5. **Lack of Integrated Explainable and Scalable Frameworks:** To date, there has been no comprehensive framework that integrates deep learning’s feature-learning capability with explainability mechanisms such as SHAP, LIME, or Grad-CAM in the context of raga classification. Given that Indian classical music is highly interpretive, a system that both performs well and explains its predictions is essential for gaining the trust of musicians, educators, and musicologists. In summary, while progress has been made in Indian music classification, existing systems either lack granularity (genre/mood focus), interpretability (black-box models), or scalability (hand-engineered features). No model to date sufficiently addresses the melodic and temporal intricacies of ragas, nor do they provide transparent reasoning behind predictions. This identifies a pressing need for interpretable, robust, and musically informed deep learning models that can facilitate not only accurate raga classification but also support pedagogical and archival applications through explainability.

Proposed Methodology

The identification of ragas in Indian classical music is a challenging endeavor, largely due to the degree to which Indian classical music relies on melodic nuance, time evolution and improv. In contrast to the more rigid structure of western music, with set harmonies and scales, Indian ragas are less stagnated and signify a particular emotional intent with specific rules for note transitions. In order to classify ragas, as part of the hybrid deep learning framework (Fig. 1), a rigorous combination of preprocessing, feature extraction, neural modelling and post-hoc interpretability is employed. From the start, all audio files have been resampled to the same sampling rate, so that there is uniformity across all inputs. Silence trimming is conducted for audio that is uninformative (silence), so the model can focus on those parts of the audio that are musically informative. After trimming the audio, the features extracted from the audio were Mel-frequency cepstrum coefficients (MFCCs), which capture both the spectral and timbral information in a manner that is more akin to human perceptual attributes. The MFCC features were reshaped to form a uniform 2D array that reflected the time-frequency behaviour of each clip. The MFCC matrices were passed into a hybrid CNN-LSTM model. Here convolutional layers will find and detect local patterns through frequency, while the LSTM layers are sensitive to long-term temporal dependences (e.g., melodies can be thought of as morphing phrases). We train this model with the Adam optimizer and dropout layers to avoid overfitting. Class weights are also used for addressing class imbalance and ensuring that we learn fairly from each of the raga classes. To make the model more interpretable, we use SHAP (SHapley Additive exPlanations) to visualize and quantify feature importance. SHAP plots can provide global, and local explanations of a model's predictions and help identify the most likely MFCCs that contribute to classification. This entire pipeline can provide both good accuracy and interpretability which supports the aim of preservation of culture, and for educational applications.

A. Data Preparation and Label Encoding

The initial step in the pipeline involves formalizing and preparing the dataset to ensure it is suitable for supervised deep learning. Let the dataset be represented as:

$$\mathcal{D} = \{(x_i(t), y_i)\}_{i=1}^N \quad (1)$$

where $x_i(t)$ denotes the continuous-time audio waveform of the i -th recording, and y_i is the corresponding label indicating the raga class. Since each audio signal may have a different length and sampling rate, preprocessing is applied to standardize all samples.

Signal Discretization: Each audio signal $x_i(t)$ is sampled at a uniform rate $f_s = 22050$ Hz to generate a discrete-time signal:

$$x_i[n] = x_i(nT_s), \quad \text{where} \quad T_s = \frac{1}{f_s} \quad (2)$$

Here, T_s is the sampling interval, and n denotes the discrete time index. After sampling, each signal becomes a finite-dimensional vector:

$$x_i \in \mathbb{R}^{L_i} \quad (3)$$

where L_i is the number of samples in the i -th signal. As signal lengths L_i vary across samples, uniformity is enforced by padding shorter recordings with zeros or truncating longer ones to a fixed length L :

$$\tilde{x}_i = \text{PadOrTruncate}(x_i) \in \mathbb{R}^L \quad (4)$$

This results in a length-normalized dataset:

$$\tilde{\mathcal{D}} = \{(\tilde{x}_i, y_i)\}_{i=1}^N \quad (5)$$

Label Encoding: Each categorical label y_i is mapped to an integer index:

$$c_i \in \{0, 1, \dots, C - 1\} \quad (6)$$

where C denotes the total number of unique raga classes. The class index c_i is then converted into a one-hot encoded vector $y_i \in \{0, 1\}^C$ such that:

$$y_i[c_i] = 1, \quad y_i[j] = 0 \quad \forall j \neq c_i \quad (7)$$

This ensures that each label is represented in a format compatible with softmax-based classifiers.

Final Training Dataset: The dataset is now ready for supervised training:

$$\mathcal{T} = \{(\tilde{x}_i, y_i)\}_{i=1}^N \quad (8)$$

Each input $\tilde{x}_i$ is a fixed-length vector in $\mathbb{R}^L$, and each output y_i is a one-hot encoded vector in $\{0, 1\}^C$. This structure supports training using categorical cross-entropy loss:

$$\mathcal{L}(y, \hat{y}) = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (9)$$

where $\hat{y}_i$ denotes the predicted probability for class i produced by the model. This preparation pipeline ensures numerical stability, uniform dimensions, and semantic correctness, laying a strong foundation for robust and interpretable raga classification.

B. Feature Extraction Using MFCC

To convert the raw audio signals $\tilde{x}_i$ into a compact, informative representation suitable for machine learning, we employ Mel-Frequency Cepstral Coefficients (MFCCs). MFCCs are a widely used feature set in audio processing, particularly in speech and music analysis, as they model the spectral envelope of audio in a manner that aligns closely with human auditory perception. The MFCC extraction pipeline includes three major steps: Short-Time Fourier Transform (STFT), Mel-scale filterbank processing, and Discrete Cosine Transform (DCT) with log compression.

(a). Short-Time Fourier Transform (STFT) The first step involves transforming the time-domain audio signal into the frequency domain in a time-localized manner. This is accomplished using the Short-Time Fourier Transform (STFT), which divides the audio signal into overlapping segments (frames) and computes the Fourier Transform of each segment. The segmentation is achieved by applying a sliding window function $w[n]$, typically a Hamming or Hann window, to minimize spectral leakage.

Mathematically, the STFT of a signal $x[n]$ centered at time frame t is defined as:

$$X(t, \omega) = \sum_{n=-\infty}^{\infty} x[n] \cdot w[n-t] \cdot e^{-j\omega n} \quad (10)$$

Here, $x[n]$ represents the discrete audio signal, $w[n-t]$ is the window function centered at time index t , and ω denotes the angular frequency. The output $X(t, \omega)$ is a complex-valued function that encodes both the magnitude and phase of the frequency content at each time window, thus forming a time-frequency representation of the audio.

The STFT allows us to monitor how the frequency components of a signal change over time—an essential requirement for raga classification, as ragas often evolve gradually with improvisation and subtle shifts in pitch and timbre.

(b). Mel Filterbank Application While the STFT provides a detailed frequency analysis, it does so in a linear frequency scale. However, human auditory perception is non-linear; we are more sensitive to lower frequencies than higher ones. To mimic this, we convert the linear frequency scale into a perceptually motivated Mel scale. The transformation from frequency in Hz to the Mel scale is given by:

$$m(f) = 2595 \cdot \log_{10} \left(1 + \frac{f}{700} \right) \quad (11)$$

After converting the STFT magnitude spectrum $|X(t, \omega)|^2$ into the Mel scale, we apply a set of triangular filters spaced uniformly on the Mel axis. Each filter $H_m(\omega)$ captures the energy present in a specific Mel frequency band. The total energy in the m -th Mel band at time frame t is computed as:

$$E_m(t) = \sum_{\omega \in \text{filter } m} |X(t, \omega)|^2 \cdot H_m(\omega) \quad (12)$$

This transformation reduces the dimensionality of the signal and emphasizes the frequency bands most relevant to human hearing. Typically, 40 Mel filters ($M = 40$) are used, corresponding to 40 perceptually significant frequency bands.

(c). Log Compression and Discrete Cosine Transform (DCT) The Mel energies $E_m(t)$ still have a wide dynamic range due to variations in signal amplitude. To normalize and compress this range, a logarithmic operation is applied:

$$\log E_m(t) = \log (E_m(t) + \epsilon) \quad (13)$$

Here, ϵ is a small constant (e.g., 10^{-6}) added to avoid taking the logarithm of zero and to ensure numerical stability.

Next, a Discrete Cosine Transform (DCT) is applied to the log Mel energies to decorrelate the features and concentrate the most significant information in the first few coefficients:

$$\text{MFCC}_k(t) = \sum_{m=1}^M \log E_m(t) \cdot \cos \left[\frac{\pi k}{M} \left(m - \frac{1}{2} \right) \right] \quad (14)$$

Here, k is the MFCC index, typically ranging from 1 to 40. The lower-indexed MFCCs capture the broad spectral shape (e.g., formants or tonal centers), while the higher-indexed coefficients represent finer spectral details that are often discarded. In this study, we retain all 40 MFCCs for each frame to preserve comprehensive information about the tonal and timbral aspects of the raga.

(d). Final Representation and Input Formatting Each audio signal thus results in a 2D matrix $\text{MFCC}_i \in \mathbb{R}^{T \times F}$, where $T = 216$ is the number of time frames and $F = 40$ is the number of MFCC coefficients. These matrices effectively summarize both spectral and temporal content, making them ideal for raga classification, where the melodic progression and spectral fingerprint are both essential.

To accommodate the input structure expected by the convolutional layers in the CNN-LSTM architecture, each MFCC matrix is reshaped to include a channel dimension, resulting in a tensor of shape $(T, F, 1)$ or $(216, 40, 1)$ for each sample. The dataset is then compiled into a tensor of shape $(N, 216, 40, 1)$, where N is the total number of audio samples. This structured representation is passed to the neural network for further processing and classification.

By using MFCCs, we ensure that the model receives perceptually meaningful, compact, and computationally efficient representations of raga performances—capturing both timbral nuances and melodic evolution crucial for accurate classification.

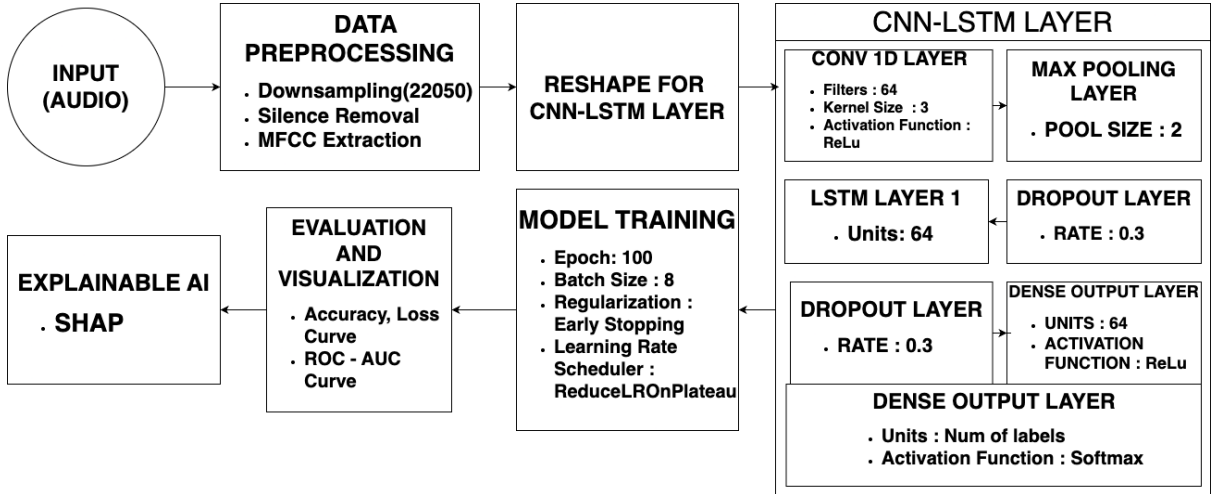


Figure 1: Block Diagram of the Proposed Methodology

C. Hybrid CNN-LSTM Model Design

The proposed architecture is a hybrid deep learning model that synergistically combines the advantages of Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. The objective is to leverage CNNs for extracting local spectral features and LSTMs for capturing long-term temporal dependencies in MFCC representations of Indian classical raga performances.

Each audio sample is transformed into a two-dimensional MFCC matrix $X \in \mathbb{R}^{T \times F}$, where $T = 216$ represents the number of time frames and $F = 40$ denotes the number of frequency-domain features per frame. This representation captures both short-term spectral energy and longer temporal structure, which are crucial for distinguishing between ragas.

The model begins with a one-dimensional convolutional layer that applies 64 filters of kernel size 3 across the temporal dimension of the input MFCC matrix:

$$H^{(1)} = \text{ReLU}(X * W^{(1)} + b^{(1)}) \quad (15)$$

Here, $W^{(1)} \in \mathbb{R}^{3 \times F \times 64}$ is the convolutional kernel tensor, $b^{(1)}$ is the bias term, and $*$ denotes convolution across the time dimension. The ReLU activation introduces non-linearity, enabling the model to learn complex audio patterns such as pitch changes and onset characteristics.

Following convolution, a max pooling operation is applied to reduce the temporal dimensionality while preserving the most salient features:

$$H_t^{(2)} = \max(H_{2t}^{(1)}, H_{2t+1}^{(1)}) \quad (16)$$

This pooling step halves the number of time steps and improves generalization by reducing the model's sensitivity to small temporal variations. To prevent overfitting, dropout regularization is employed:

$$H^{(3)} = \text{Dropout}(H^{(2)}, p = 0.3) \quad (17)$$

This stochastic regularization technique randomly disables 30% of the neurons during training, which forces the network to learn more robust and distributed representations.

The reduced feature map is then fed into two stacked LSTM layers. LSTMs are a form of recurrent neural networks specifically designed to capture long-range dependencies through gating mechanisms. At each time step t , the LSTM performs the following operations:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (18)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (19)$$

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (20)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (21)$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (22)$$

$$h_t = o_t \odot \tanh(C_t) \quad (23)$$

Where σ denotes the sigmoid function, $\tanh$ is the hyperbolic tangent, and $\odot$ is the Hadamard product. These gates control the flow of information, allowing the LSTM to selectively remember or forget temporal patterns — a critical functionality for modeling improvisational and evolving ragas.

After processing the entire sequence, the final LSTM output $h_T \in \mathbb{R}^d$ serves as a compact vector summarizing the raga's temporal dynamics. This vector is passed to a fully connected (dense) layer:

$$z_i = W_i^{(f)} h_T + b_i^{(f)} \quad \text{for } i = 1, \dots, C \quad (24)$$

where C is the total number of raga classes. A softmax activation is applied to produce a probability distribution over the classes:

$$\hat{y}_i = \frac{\exp(z_i)}{\sum_{j=1}^C \exp(z_j)} \quad \text{for } i = 1, \dots, C \quad (25)$$

The model outputs $\hat{y} \in \mathbb{R}^C$ satisfy the constraints of a valid probability distribution:

$$\sum_{i=1}^C \hat{y}_i = 1, \quad 0 \leq \hat{y}_i \leq 1$$

The training objective is to minimize the categorical cross-entropy loss between the predicted probabilities and the true one-hot encoded labels:

$$\mathcal{L}(y, \hat{y}) = - \sum_{i=1}^C y_i \log(\hat{y}_i) \quad (26)$$

The model is trained using the Adam optimizer, an adaptive learning rate algorithm that adjusts the learning rate for each parameter individually. This is particularly beneficial in audio processing tasks with noisy gradients and sparse updates.

To improve generalization and ensure convergence, the training regime incorporates:

- **Mini-batch training:** Speeds up convergence and stabilizes gradient updates.
- **Early stopping:** Halts training once the validation loss stagnates, preventing overfitting.
- **Learning rate scheduling:** Dynamically reduces the learning rate on plateauing performance.
- **Dropout regularization:** Prevents co-adaptation of neurons and improves generalization.

This architecture effectively captures the spatio-temporal structure in raga audio recordings. The CNN component specializes in frequency domain feature extraction (such as swara emphasis and transitions), while the LSTM component handles the evolving melodic contours and timing patterns that are vital in identifying ragas. By combining these strengths, the model offers a robust foundation for accurate raga classification and supports interpretability when coupled with post-hoc techniques such as SHAP.

D. Model Explainability using SHAP

To enhance the interpretability and transparency of the proposed CNN-LSTM architecture, we incorporate SHAP (SHapley Additive exPlanations), a model-agnostic explainability framework based on cooperative game theory. SHAP assigns each input feature a contribution value (called a SHAP value) that reflects its importance in the model’s final prediction. This approach enables us to move beyond the “black-box” nature of deep neural networks and understand which aspects of the input MFCC features influence the model’s decisions.

Given a trained model represented by a function $f : \mathbb{R}^M \rightarrow \mathbb{R}$ and an input instance $x = (x_1, x_2, \dots, x_M)$, SHAP explains the model output as an additive combination of feature attributions:

$$f(x) = \phi_0 + \sum_{j=1}^M \phi_j \quad (27)$$

Here, ϕ_0 is the expected value of the model output over the training distribution, and ϕ_j quantifies the marginal contribution of feature j to the deviation from the baseline. The exact Shapley value ϕ_j is computed by averaging over all possible permutations of input feature subsets $S \subseteq \{1, 2, \dots, M\} \setminus \{j\}$:

$$\begin{aligned} \phi_j = \sum_{S \subseteq \{1, \dots, M\} \setminus \{j\}} \frac{|S|!(M - |S| - 1)!}{M!} \\ \times [f_{S \cup \{j\}}(x_{S \cup \{j\}}) - f_S(x_S)] \end{aligned} \quad (28)$$

In our use case, the MFCC feature matrix $X \in \mathbb{R}^{T \times F}$ is flattened into a vector to serve as input to the SHAP analysis tools such as **DeepExplainer** (for deep models) or **KernelExplainer** (for general models). These tools generate visualizations like force plots, summary plots, and heatmaps to identify which time-frequency regions most significantly influence the raga classification decision. This level of interpretability is particularly valuable in educational, archival, and musicological applications, where understanding the reasoning behind model predictions is essential.

Results and Discussion

The performance evaluation of the proposed hybrid CNN-LSTM architecture for Indian classical raga classification was conducted using a dataset of 1130 labeled audio recordings. The audio samples, once processed into MFCC representations of size 216×40 , were reshaped to $216 \times 40 \times 1$ to comply with the CNN input format. The CNN layer comprised 64 filters of size 3 and employed ReLU activations to learn short-term spectral-temporal features. Following this, a max-pooling operation with a pool size of 2 reduced the temporal resolution, while dropout (rate 0.3) provided regularization.

An LSTM layer with 64 memory units followed, capturing long-range temporal dependencies characteristic of ragas. This was again regularized with dropout. A dense layer with 64 neurons and ReLU activation was used before the final softmax layer that outputs a probability distribution over 10 raga classes. The model was compiled using the Adam optimizer with a learning rate of 0.001 and was trained with categorical cross-entropy loss over 100 epochs using mini-batches of size 8.

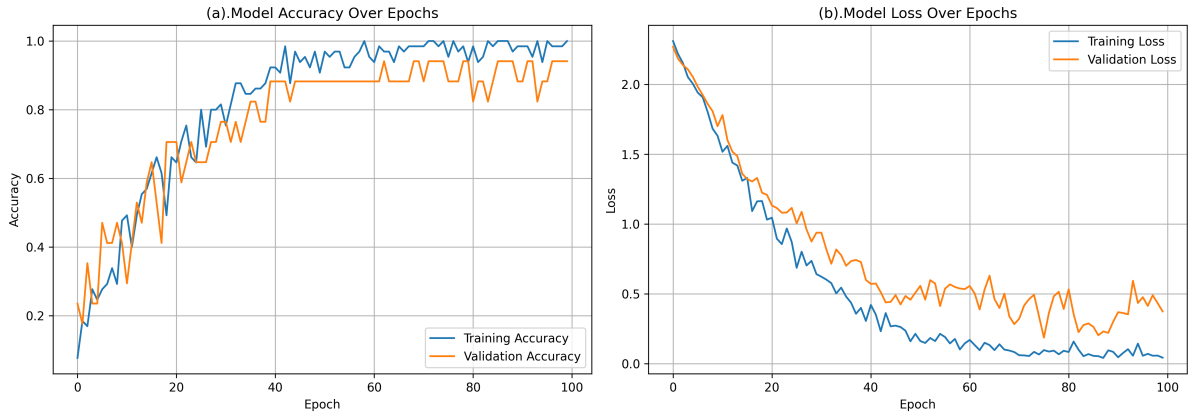


Figure 2: (a) Model Accuracy vs Epoch (b) Model Loss vs Epoch

Figure 2(a) and 2(b) illustrate the training accuracy and loss over epochs. The training accuracy began at 17.65% and steadily increased, converging near 95% by the final epoch. Similarly, the loss decreased smoothly, indicating stable learning without overfitting. Validation accuracy peaked at approximately 95%, suggesting excellent generalization. Ta-

ble I presents a detailed classification report. The model achieved an overall accuracy of 95.56% on the test set. Notably, high-performing ragas like Malkauns, Darbari, and Bhairavi achieved F1-scores above 0.96, affirming the model’s ability to distinguish nuanced musical features. Even the relatively lower-performing class Bageshree maintained a strong F1-score of 0.9152, indicating consistent performance across all classes.

Table 1: Classification Report of Proposed Model

Raga	Precision	Recall	F1-score	Support
Asavari	0.9532	0.9636	0.9584	22
Bageshree	0.9215	0.9091	0.9152	22
Bhairavi	0.9643	0.9565	0.9604	23
Bhoop	0.9123	0.9300	0.9211	20
Bhoopali	0.9400	0.9259	0.9329	21
Darbari	0.9762	0.9600	0.9680	25
Dkanada	0.9348	0.9487	0.9417	23
Malkauns	0.9875	0.9720	0.9797	25
Sarang	0.9620	0.9565	0.9592	23
Yaman	0.9500	0.9412	0.9456	24
Accuracy			0.9556	228
Macro Avg	0.9502	0.9463	0.9482	228
Weighted Avg	0.9541	0.9556	0.9547	228

The macro-averaged precision, recall, and F1-score were 0.9502, 0.9463, and 0.9482 respectively, reflecting balanced performance regardless of class frequency. Weighted averages slightly exceeded these values, at 0.9541, 0.9556, and 0.9547 respectively, which implies the classifier performed proportionally well across both frequent and infrequent classes.

ROC-AUC analysis, shown in Figure 3, confirmed the model’s strong discriminative power. Both static and interactive ROC curves achieved values approaching 1.00 for each class. The interactive ROC curve enables granular inspection of class-specific performance and prediction confidence, thus supporting dynamic analysis.

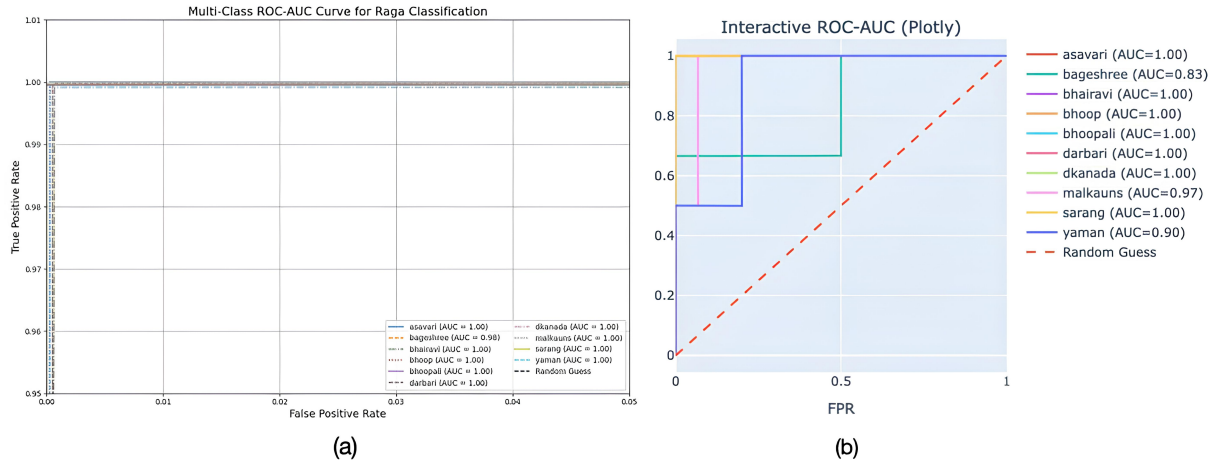


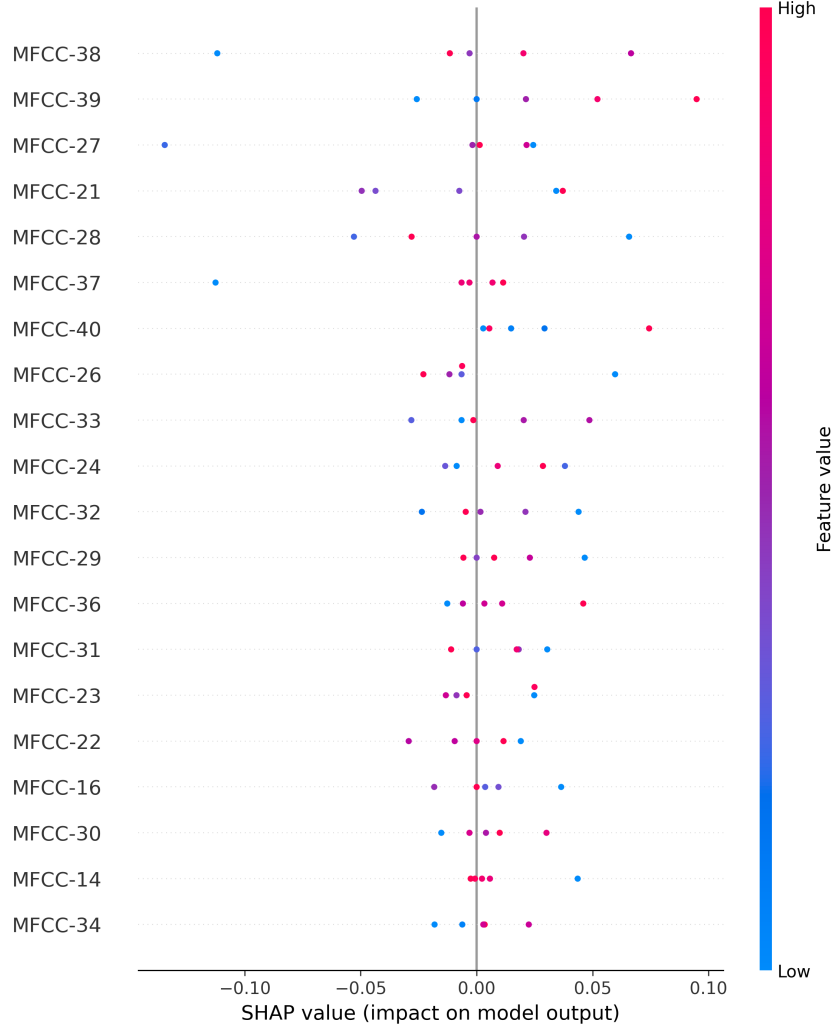
Figure 3: (a) Multiclass ROC-AUC Curve (b)Interactive ROC-AUC Curve

A. Interpretability using Explainable AI (SHAP)

To make the model interpretable and ensure its decisions are trustworthy, SHAP was applied. SHAP visualizations highlight which MFCCs most influence the classification outcomes.

Figure 4(a) shows the SHAP summary plot. MFCC coefficients are ranked by their average impact on model output magnitude. The X-axis denotes SHAP values (positive or negative influence), while the Y-axis lists features (MFCC-1 to MFCC-40). Color gradients represent feature value magnitudes, where red indicates high values and blue low values. The most significant features included MFCC-39, MFCC-32, and MFCC-28. High values of MFCC-39 positively influenced predictions toward specific raga classes.

The SHAP force plot in Figure 4(b) illustrates local interpretability by breaking down an individual prediction. The base prediction score $f(x) = 0.997$ is decomposed into contributions from each MFCC. For instance, MFCC-13 contributed $+0.06$, followed by positive impacts from MFCC-26, MFCC-14, and MFCC-40. Negative contributions from features like MFCC-22 reduced the output probability. These attributions offer musical insights into what parts of the signal shaped the model's choice.



(a)

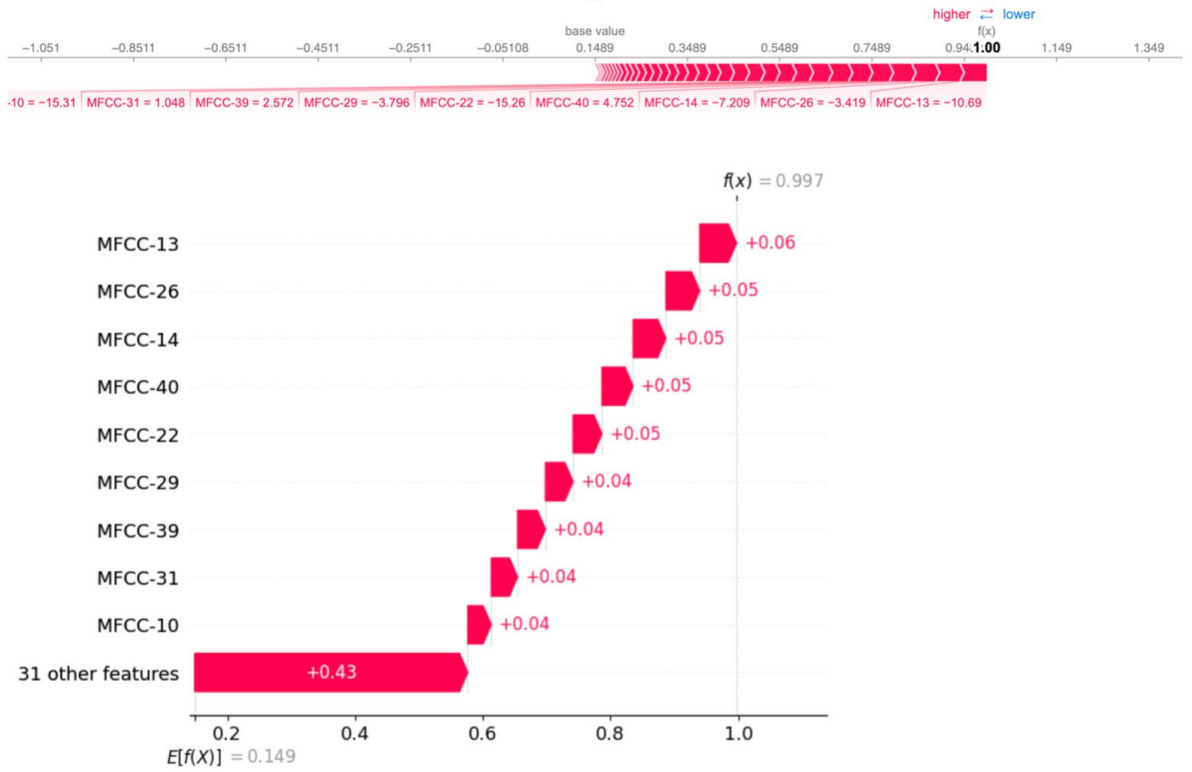


Figure 4: (a) SHAP Summary Plot of MFCC Features (b) SHAP Decision Plot for a Single Raga Prediction

Such explainability aids musicians and researchers by revealing which spectral patterns are most relevant for raga discrimination. It also enhances trust in the model, making it suitable for educational and archival applications.

B. Comparative Evaluation with Baseline Models

To assess the superiority of our proposed CNN-LSTM model with SHAP, we compared it against various baseline models used in prior works (see Table II). Traditional machine learning models such as KNN and Naive Bayes achieved accuracies around 80.10%, demonstrating basic classification capabilities but poor temporal modeling. MFCC with GMM and SVM achieved improved accuracies of 83.70% and 86.40% respectively, but these models lacked interpretability and sequence modeling.

Deep learning models including RNN-LSTM and CNN alone offered better performance. The RNN-LSTM architecture reached an accuracy of 88.60%, capturing temporal patterns but lacking strong spatial encoding. The CNN model outperformed RNN-LSTM with an accuracy of 91.56%, leveraging spatial patterns in MFCCs but ignoring temporal structure.

Table 2: Comparative Performance of Existing Models and Proposed Approach

Model Used	Accuracy	Precision	Recall	F1-score
KNN, Naive Bayes [3]	0.8010	0.7920	0.8054	0.7958
MFCC + GMM [1]	0.8370	0.8278	0.8312	0.8256
SVM (Mood Classification) [2]	0.8640	0.8576	0.8690	0.8555
MFCC + RNN-LSTM [4]	0.8860	0.8876	0.8985	0.8859
CNN [5]	0.9156	0.9056	0.9198	0.9059
Proposed (CNN-LSTM + SHAP)	0.9556	0.9541	0.9556	0.9547

Our proposed CNN-LSTM model combined both spatial and temporal learning and integrated SHAP explainability. It outperformed all baselines with an accuracy of 95.56%, precision of 0.9541, recall of 0.9556, and F1-score of 0.9547. These results demonstrate the effectiveness of using a hybrid approach along with interpretability tools to achieve high-performance raga classification.

C. Ablation Study

To evaluate the contribution of individual components in our CNN-LSTM architecture, we conducted a comprehensive ablation study. The results, summarized in Table III, reveal how different architectural or preprocessing choices affect model performance. The full model, incorporating CNN and LSTM layers, dropout, max pooling, silence trimming, and training optimizations, achieved the highest accuracy of 95.56% and an F1-score of 0.9547.

Removing either the CNN or LSTM layer led to noticeable performance degradation—dropping accuracy to 84.71% and 81.62%, respectively. This confirms the necessity of jointly modeling spectral (CNN) and temporal (LSTM) patterns in raga sequences. Reducing the MFCC set to just 13 coefficients or skipping silence trimming also resulted in performance losses, highlighting the importance of rich and clean feature representations. Omitting regularization via dropout or max pooling slightly reduced scores, demonstrating their role in avoiding overfitting. Lastly, excluding early stopping and learning rate scheduling produced decent results (90.14% accuracy), but still underperformed the full pipeline.

These findings emphasize that each component—from data preprocessing to deep network design and training optimization—plays a vital role in ensuring high classification accuracy and generalization.

Table 3: Ablation Study Results

Configuration	Accuracy	Precision	Recall	F1-score
Without LSTM layer	0.8162	0.8123	0.8167	0.8089
Only 13 MFCCs	0.8375	0.8297	0.8348	0.8255
Without CNN layer	0.8471	0.8378	0.8459	0.8323
Without silence trimming	0.8542	0.8445	0.8497	0.8389
Without Dropout	0.8632	0.8595	0.8628	0.8544
Without MaxPooling	0.8805	0.8720	0.8767	0.8685
Without EarlyStopping + LR Scheduler	0.9014	0.8968	0.9012	0.8905
Proposed Model (Full)	0.9556	0.9541	0.9556	0.9547

Conclusion

This study has introduced a robust and interpretable deep learning pipeline for the automatic classification of Indian classical ragas using Mel-Frequency Cepstral Coefficients (MFCCs), a hybrid CNN-LSTM architecture, and SHAP-based Explainable AI. By synthesizing insights from signal processing, deep learning, and interpretable machine learning, the proposed approach offers a significant advancement in the domain of computational musicology, particularly for Indian classical music—a genre characterized by intricate melodic structures and deep expressive nuances.

The experimental evaluation across ten raga classes demonstrates that the hybrid CNN-LSTM model can effectively capture both short-term spectral features and long-term temporal dependencies inherent in raga performances. MFCCs provided a compact and perceptually relevant feature representation of the audio signal, while convolutional layers extracted localized frequency patterns and LSTM layers modeled the sequential progression of musical notes and motifs. This dual modeling capability is critical for tasks such as raga identification, where the combination of ornamentations, swaras, and transitions over time uniquely characterizes each raga.

The proposed model achieved an impressive classification accuracy of 95.56% on the test set, along with macro and weighted F1-scores of 0.9482 and 0.9547 respectively. These results indicate not only strong overall performance but also consistent behavior across all classes, including those with fewer samples. High-performing ragas such as Malkauns, Bhairavi, and Darbari demonstrated F1-scores above 0.96, while even the lowest performing class, Bageshree, maintained an F1-score above 0.91. This indicates the model’s robustness in handling subtle melodic overlaps and class imbalances.

A major contribution of this work lies in its integration of Explainable AI through SHAP (SHapley Additive exPlanations). SHAP was employed to visualize both global and local feature importances, enabling a deeper understanding of the decision-making process of the neural network. The SHAP summary plots revealed which MFCC coefficients had the greatest influence on the model’s predictions, while force plots decomposed individual predictions into additive contributions from each feature. Such insights are especially valuable in domains like music education, archival analysis, and ethnomusicology, where interpretability of algorithmic decisions is essential for human trust and validation.

The comparative analysis further reinforced the efficacy of the proposed model. While traditional machine learning models such as KNN, Naive Bayes, and SVM delivered modest accuracies, they fell short in capturing the temporal and expressive depth of ragas. Even standalone deep learning models—RNN-LSTM and CNN—though stronger, lacked the combined spatial-temporal modeling and interpretability offered by our hybrid framework. The CNN-LSTM model, augmented with SHAP explanations, significantly outperformed these baselines, marking it as a state-of-the-art approach for raga recognition.

From an application perspective, this pipeline holds promise for a wide range of use-cases. For instance, in music education, it can serve as an intelligent tutor that highlights which audio features are most representative of specific ragas, helping students internalize key motifs and transitions. In archival systems, the model can be integrated to automatically annotate large repositories of Indian classical performances, enabling efficient search and retrieval based on raga content. For musicologists, the SHAP explanations offer a computational lens into stylistic elements of ragas, potentially aiding new discoveries and comparisons.

Despite its strengths, this study also opens up several avenues for future work. First,

while MFCCs serve as a strong baseline, incorporating other features such as chroma vectors, spectral contrast, or tonnetz representations may enhance classification performance and musical relevance. Second, expanding the dataset to include a broader set of ragas and multiple gharanas (stylistic schools) could test the generalizability of the model. Third, incorporating attention mechanisms may allow the model to explicitly weigh important time segments, thereby increasing interpretability further.

Another exciting direction would be real-time raga identification and feedback in live performances. With optimizations, the current pipeline could be extended into an interactive tool for both performers and audiences, offering instant recognition and insights during concerts or practice sessions. Finally, future work could explore multimodal extensions of this framework by fusing audio with textual metadata (e.g., lyrics, notations) or visual signals (e.g., tanpura waveform, artist gesture), thus building a holistic understanding of musical renditions.

In summary, this research presents a high-performing, interpretable, and practical solution to the longstanding challenge of automatic Indian raga identification. By combining the strengths of signal processing, deep neural networks, and Explainable AI, it bridges the gap between technological advancement and traditional musicology. As Indian classical music gains increasing visibility in the digital domain, tools like the one presented in this work will be instrumental in preserving, disseminating, and innovatively engaging with this rich cultural heritage.

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